

THE COMPLETE GUIDE TO

ALL-LECTURE CHEAT SHEETS

A scan-friendly field reference for every platform lecture: 2 introductory lectures plus all 365 numbered topics.



52 WEEKS	365 NUMBERED TOPICS	ZERO TO HERO
367 lecture cards	Professional resource edition	Concepts + practice + evidence

HOW TO USE THIS BOOK

Each card is designed for a two-minute review before or after a lecture. The core idea states the durable principle; the evidence check identifies what a practitioner should inspect; the quick action converts theory into a command, formula, or decision record.

The cards intentionally avoid version-specific flags where those change quickly. Use the maintained official references in the companion handbook when completing a lab.

OUTCOMES

SLOs - safety - cost - user value

AI WORKLOAD

training - inference - retrieval - agents

PLATFORM

pipelines - serving - orchestration - registry

RUNTIME

frameworks - containers - drivers - libraries

FOUNDATION

compute - network - storage - facilities

THE THREE-PASS REVIEW

- Pass 1 - Recognize: define the concept without naming a vendor.
- Pass 2 - Apply: state the workload condition that changes your decision.
- Pass 3 - Prove: name the metric, test, or artifact that validates the decision.

LECTURE 1 | COURSE INTRO

Certificate of Completion

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 2 | COURSE INTRO

Introduction to The Complete Guide to AI Infrastructure: Zero to Hero

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 3 | TOPIC 1 | WEEK 1

What Is AI Infrastructure? The Engine Behind AI

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 4 | TOPIC 2 | WEEK 1

CPU vs GPU vs TPU - Computing Power for AI

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 5 | TOPIC 3 | WEEK 1

Training vs Inference - Two Faces of AI Workloads

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 6 | TOPIC 4 | WEEK 1

AI Infrastructure Layers - Hardware, Software, Ops

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 7 | TOPIC 5 | WEEK 1

Case Studies: Infrastructure Behind ChatGPT, DALL-E

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 8 | TOPIC 6 | WEEK 1

Industry Landscape: Cloud Providers & AI Chips

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 9 | TOPIC 7 | WEEK 1

Lab - Spin Up Your First AI VM

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 10 | TOPIC 8 | WEEK 2

Why Linux Dominates AI Infrastructure

Core idea. Use the operating system as an observable substrate: know where state lives, who can access it, and which process consumes resources.

Evidence check. Inspect identity, paths, permissions, processes, open ports, disk, memory, and logs before changing state.

Quick action. `id; pwd; df -h; free -h 2>/dev/null || vm_stat; ps aux | head`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 11 | TOPIC 9 | WEEK 2

Navigating the Linux Shell - Bash Basics

Core idea. Use the operating system as an observable substrate: know where state lives, who can access it, and which process consumes resources.

Evidence check. Inspect identity, paths, permissions, processes, open ports, disk, memory, and logs before changing state.

Quick action. `id; pwd; df -h; free -h 2>/dev/null || vm_stat; ps aux | head`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 12 | TOPIC 10 | WEEK 2

Filesystems, Directories, and Permissions

Core idea. Use the operating system as an observable substrate: know where state lives, who can access it, and which process consumes resources.

Evidence check. Inspect identity, paths, permissions, processes, open ports, disk, memory, and logs before changing state.

Quick action. `id; pwd; df -h; free -h 2>/dev/null || vm_stat; ps aux | head`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 13 | TOPIC 11 | WEEK 2

Package Managers - apt, yum, pip

Core idea. Use the operating system as an observable substrate: know where state lives, who can access it, and which process consumes resources.

Evidence check. Inspect identity, paths, permissions, processes, open ports, disk, memory, and logs before changing state.

Quick action. `id; pwd; df -h; free -h 2>/dev/null || vm_stat; ps aux | head`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 14 | TOPIC 12 | WEEK 2

Process Management and Monitoring

Core idea. Use the operating system as an observable substrate: know where state lives, who can access it, and which process consumes resources.

Evidence check. Inspect identity, paths, permissions, processes, open ports, disk, memory, and logs before changing state.

Quick action. `id; pwd; df -h; free -h 2>/dev/null || vm_stat; ps aux | head`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 15 | TOPIC 13 | WEEK 2

Scripting for Automation in AI Workflows

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 16 | TOPIC 14 | WEEK 2

Lab - Set Up Ubuntu for AI Development

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 17 | TOPIC 15 | WEEK 3

Introduction to Cloud for AI Workloads

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. record region + instance + accelerator + image + storage + hourly estimate

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 18 | TOPIC 16 | WEEK 3

Compute Instances - Choosing CPU vs GPU

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 19 | TOPIC 17 | WEEK 3

Networking Basics - VPCs, Firewalls, Load Balancers

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 20 | TOPIC 18 | WEEK 3

Cloud Storage - Object, Block, and File Systems

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 21 | TOPIC 19 | WEEK 3

Hands-On with AWS EC2 for AI

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 22 | TOPIC 20 | WEEK 3

Hands-On with Google Cloud GPU Instances

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 23 | TOPIC 21 | WEEK 3

Lab - Compare Cost & Performance Across Clouds

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 24 | TOPIC 22 | WEEK 4

Why Containers Are Critical for AI

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 25 | TOPIC 23 | WEEK 4

Docker Basics - Images and Containers

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 26 | TOPIC 24 | WEEK 4

Building a Container for an AI App

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 27 | TOPIC 25 | WEEK 4

Networking and Volumes in Docker

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 28 | TOPIC 26 | WEEK 4

Docker Compose for Multi-Service AI Systems

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 29 | TOPIC 27 | WEEK 4

Best Practices: Lightweight AI Containers

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 30 | TOPIC 28 | WEEK 4

Lab - Containerize a PyTorch Model

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 31 | TOPIC 29 | WEEK 5

What Is Kubernetes and Why AI Needs It

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 32 | TOPIC 30 | WEEK 5

Pods, Nodes, and Clusters Explained

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 33 | TOPIC 31 | WEEK 5

Deployments and Services for AI Apps

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 34 | TOPIC 32 | WEEK 5

ConfigMaps, Secrets, and Volumes

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 35 | TOPIC 33 | WEEK 5

Horizontal Pod Autoscaling for AI Inference

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 36 | TOPIC 34 | WEEK 5

Helm Charts for Simplified AI Deployments

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 37 | TOPIC 35 | WEEK 5

Lab - Deploy a Model on Minikube

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 38 | TOPIC 36 | WEEK 6

Data Lakes vs Data Warehouses vs Feature Stores

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 39 | TOPIC 37 | WEEK 6

Object Storage with AWS S3 and GCP GCS

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 40 | TOPIC 38 | WEEK 6

Relational vs NoSQL Databases in AI

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 41 | TOPIC 39 | WEEK 6

Streaming Data with Kafka & Pub/Sub

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. Record: `input -> transform -> output -> checkpoint -> alert`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 42 | TOPIC 40 | WEEK 6

Scaling Storage for AI Training Datasets

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 43 | TOPIC 41 | WEEK 6

Secure Data Access and Encryption

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 44 | TOPIC 42 | WEEK 6

Lab - Build a Data Ingestion Pipeline

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. Record: input -> transform -> output -> checkpoint -> alert

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 45 | TOPIC 43 | WEEK 7

Anatomy of a GPU for AI

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 46 | TOPIC 44 | WEEK 7

CUDA Basics and GPU Programming

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 47 | TOPIC 45 | WEEK 7

GPU Memory Hierarchy - Optimizing Usage

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 48 | TOPIC 46 | WEEK 7

Multi-GPU Scaling and Interconnects (NVLink)

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 49 | TOPIC 47 | WEEK 7

Multi-Instance GPU (MIG) Configurations

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 50 | TOPIC 48 | WEEK 7

Benchmarking AI Workloads on GPUs

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 51 | TOPIC 49 | WEEK 7

Lab - Run a Model on GPU with CUDA

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 52 | TOPIC 50 | WEEK 8

Why Distributed Training Is Needed

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 53 | TOPIC 51 | WEEK 8

Data Parallelism vs Model Parallelism

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 54 | TOPIC 52 | WEEK 8

PyTorch Distributed Training

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 55 | TOPIC 53 | WEEK 8

TensorFlow Multi-GPU Training

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 56 | TOPIC 54 | WEEK 8

Horovod and AllReduce Explained

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. `efficiency = single_device_time / (devices * distributed_time)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 57 | TOPIC 55 | WEEK 8

Fault Tolerance in Distributed Training

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. `efficiency = single_device_time / (devices * distributed_time)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 58 | TOPIC 56 | WEEK 8

Lab - Train ResNet Across Multiple GPUs

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 59 | TOPIC 57 | WEEK 9

Why Tracking ML Experiments Matters

Core idea. Tie each result to code, data, environment, parameters, metrics, artifacts, and an accountable lifecycle decision.

Evidence check. Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Quick action. `mlflow server --host 127.0.0.1 --port 5000`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 60 | TOPIC 58 | WEEK 9

Introduction to MLflow

Core idea. Tie each result to code, data, environment, parameters, metrics, artifacts, and an accountable lifecycle decision.

Evidence check. Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Quick action. `mlflow server --host 127.0.0.1 --port 5000`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 61 | TOPIC 59 | WEEK 9

Logging Metrics and Artifacts

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 62 | TOPIC 60 | WEEK 9

Versioning Data, Models, and Parameters

Core idea. Tie each result to code, data, environment, parameters, metrics, artifacts, and an accountable lifecycle decision.

Evidence check. Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Quick action. `mlflow server --host 127.0.0.1 --port 5000`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 63 | TOPIC 61 | WEEK 9

Weights & Biases vs MLflow Comparison

Core idea. Tie each result to code, data, environment, parameters, metrics, artifacts, and an accountable lifecycle decision.

Evidence check. Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Quick action. `mlflow server --host 127.0.0.1 --port 5000`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 64 | TOPIC 62 | WEEK 9

Automating Training Pipelines

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 65 | TOPIC 63 | WEEK 9

Lab - Track Experiments with MLflow

Core idea. Tie each result to code, data, environment, parameters, metrics, artifacts, and an accountable lifecycle decision.

Evidence check. Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Quick action. `mlflow server --host 127.0.0.1 --port 5000`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 66 | TOPIC 64 | WEEK 10

What Is MLOps and CI/CD for AI?

Core idea. Automate a gated path from change to verified artifact to controlled rollout, with rollback as a tested operation.

Evidence check. Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Quick action. Build -> test -> scan -> sign -> deploy -> verify -> promote/rollback

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 67 | TOPIC 65 | WEEK 10

GitHub Actions Basics for ML Projects

Core idea. Automate a gated path from change to verified artifact to controlled rollout, with rollback as a tested operation.

Evidence check. Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Quick action. `Build -> test -> scan -> sign -> deploy -> verify -> promote/rollback`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 68 | TOPIC 66 | WEEK 10

GitLab CI and Jenkins Pipelines

Core idea. Automate a gated path from change to verified artifact to controlled rollout, with rollback as a tested operation.

Evidence check. Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Quick action. `Build -> test -> scan -> sign -> deploy -> verify -> promote/rollback`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 69 | TOPIC 67 | WEEK 10

Testing Models Before Deployment

Core idea. Automate a gated path from change to verified artifact to controlled rollout, with rollback as a tested operation.

Evidence check. Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Quick action. `Build -> test -> scan -> sign -> deploy -> verify -> promote/rollback`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 70 | TOPIC 68 | WEEK 10

Automating Docker Builds for AI Apps

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 71 | TOPIC 69 | WEEK 10

Canary Deployments and Rollbacks

Core idea. Automate a gated path from change to verified artifact to controlled rollout, with rollback as a tested operation.

Evidence check. Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Quick action. `Build -> test -> scan -> sign -> deploy -> verify -> promote/rollback`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 72 | TOPIC 70 | WEEK 10

Lab - CI/CD Pipeline for Model Deployment

Core idea. Automate a gated path from change to verified artifact to controlled rollout, with rollback as a tested operation.

Evidence check. Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Quick action. `Build -> test -> scan -> sign -> deploy -> verify -> promote/rollback`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 73 | TOPIC 71 | WEEK 11

GPU Scheduling in Kubernetes

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 74 | TOPIC 72 | WEEK 11

StatefulSets for Data-Heavy Workloads

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 75 | TOPIC 73 | WEEK 11

Kubernetes Operators for ML

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 76 | TOPIC 74 | WEEK 11

Helm Advanced Templates for AI Apps

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 77 | TOPIC 75 | WEEK 11

Scaling AI Training with Kubeflow

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 78 | TOPIC 76 | WEEK 11

Kubernetes Security Best Practices

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 79 | TOPIC 77 | WEEK 11

Lab - Deploy MLflow on Kubernetes

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 80 | TOPIC 78 | WEEK 12

Why AI Infrastructure Costs Spike

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 81 | TOPIC 79 | WEEK 12

Using Spot Instances for AI Training

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 82 | TOPIC 80 | WEEK 12

Autoscaling AI Clusters in Cloud

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 83 | TOPIC 81 | WEEK 12

Monitoring Resource Utilization

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 84 | TOPIC 82 | WEEK 12

Storage Cost Optimization Strategies

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 85 | TOPIC 83 | WEEK 12

Multi-Tenant Cost Allocation in Teams

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 86 | TOPIC 84 | WEEK 12

Lab - Optimize Cloud AI Workload Costs

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 87 | TOPIC 85 | WEEK 13

Fundamentals of Data Center Networking

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 88 | TOPIC 86 | WEEK 13

Software Defined Networking (SDN) for AI

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 89 | TOPIC 87 | WEEK 13

Infiniband and High-Speed Interconnects

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 90 | TOPIC 88 | WEEK 13

Load Balancing for AI Inference

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 91 | TOPIC 89 | WEEK 13

Network Bottlenecks in Distributed Training

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 92 | TOPIC 90 | WEEK 13

Security in Networked AI Systems

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 93 | TOPIC 91 | WEEK 13

Lab - Configure Load Balancer for AI API

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 94 | TOPIC 92 | WEEK 14

From Training to Serving - The Deployment Gap

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 95 | TOPIC 93 | WEEK 14

REST vs gRPC for Model APIs

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 96 | TOPIC 94 | WEEK 14

TensorFlow Serving for AI Models

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 97 | TOPIC 95 | WEEK 14

TorchServe for PyTorch Models

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 98 | TOPIC 96 | WEEK 14

Deploying Models with FastAPI

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 99 | TOPIC 97 | WEEK 14

Scaling Model Serving with Kubernetes

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Serve an Image Classifier with FastAPI

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

NVIDIA TensorRT Optimization

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Triton Inference Server Basics

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 103 | TOPIC 101 | WEEK 15

Batch Inference vs Online Inference

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 104 | TOPIC 102 | WEEK 15

Caching for Fast Inference

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 105 | TOPIC 103 | WEEK 15

Multi-Model Serving Strategies

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 106 | TOPIC 104 | WEEK 15

A/B Testing Model Endpoints

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 107 | TOPIC 105 | WEEK 15

Lab - Deploy a Model on Triton Server

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 108 | TOPIC 106 | WEEK 16

Why Monitoring AI Systems Matters

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 109 | TOPIC 107 | WEEK 16

GPU Monitoring with DCGM

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 110 | TOPIC 108 | WEEK 16

Metrics Collection with Prometheus

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 111 | TOPIC 109 | WEEK 16

Visualization Dashboards with Grafana

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 112 | TOPIC 110 | WEEK 16

Tracing AI Requests with OpenTelemetry

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 113 | TOPIC 111 | WEEK 16

Building Alerts for AI System Failures

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 114 | TOPIC 112 | WEEK 16

Lab - Monitor GPU Cluster with Prometheus

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 115 | TOPIC 113 | WEEK 17

What Is Concept Drift vs Data Drift?

Core idea. Compare live data and outcomes with a justified reference, then route meaningful change into diagnosis and response.

Evidence check. Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.

Quick action. `reference window + current window + threshold + owner + action`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 116 | TOPIC 114 | WEEK 17

Why Drift Destroys AI Performance

Core idea. Compare live data and outcomes with a justified reference, then route meaningful change into diagnosis and response.

Evidence check. Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.

Quick action. `reference window + current window + threshold + owner + action`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 117 | TOPIC 115 | WEEK 17

Tools for Drift Detection (EvidentlyAI)

Core idea. Compare live data and outcomes with a justified reference, then route meaningful change into diagnosis and response.

Evidence check. Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.

Quick action. `reference window + current window + threshold + owner + action`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Real-Time Drift Monitoring Pipelines

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Human-in-the-Loop Drift Evaluation

Core idea. Compare live data and outcomes with a justified reference, then route meaningful change into diagnosis and response.

Evidence check. Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.

Quick action. `reference window + current window + threshold + owner + action`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Mitigation Strategies - Retraining & Rebalancing

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 121 | TOPIC 119 | WEEK 17

Lab - Build a Drift Detection Pipeline

Core idea. Compare live data and outcomes with a justified reference, then route meaningful change into diagnosis and response.

Evidence check. Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.

Quick action. `reference window + current window + threshold + owner + action`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 122 | TOPIC 120 | WEEK 18

Security Risks in AI Infrastructure

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. `who can do what to which resource under which condition?`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 123 | TOPIC 121 | WEEK 18

Identity and Access Management (IAM)

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. `who can do what to which resource under which condition?`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Secrets Management for AI Systems

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Data Encryption at Rest and In Transit

Core idea. Design each network path around trust boundaries, bandwidth, latency, connection behavior, and failure handling.

Evidence check. Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Quick action. `curl -fsS -w '%{http_code} %{time_total}\n' -o /dev/null http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Model Theft and Adversarial Attacks

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Compliance Standards (GDPR, HIPAA, SOC2)

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Secure a Model Endpoint with Authentication

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Why AI Systems Fail in Production

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 130 | TOPIC 128 | WEEK 19

Fault Tolerance in AI Inference

Core idea. Turn failure assumptions into redundancy, bounded recovery objectives, tested failover, and graceful degradation.

Evidence check. Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.

Quick action. `availability target -> error budget -> controls -> test evidence`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 131 | TOPIC 129 | WEEK 19

Redundancy and Failover for AI APIs

Core idea. Turn failure assumptions into redundancy, bounded recovery objectives, tested failover, and graceful degradation.

Evidence check. Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.

Quick action. `availability target -> error budget -> controls -> test evidence`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 132 | TOPIC 130 | WEEK 19

Designing High-Availability AI Clusters

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Auto-Healing Infrastructure with Kubernetes

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Chaos Engineering for AI Systems

Core idea. Turn failure assumptions into redundancy, bounded recovery objectives, tested failover, and graceful degradation.

Evidence check. Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.

Quick action. `availability target -> error budget -> controls -> test evidence`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Build a HA AI Inference Cluster

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Why Enterprises Go Multi-Cloud

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Kubernetes Federation Across Clouds

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Data Replication Across Cloud Providers

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Cost Arbitrage in Multi-Cloud AI

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Disaster Recovery Across Regions

Core idea. Turn failure assumptions into redundancy, bounded recovery objectives, tested failover, and graceful degradation.

Evidence check. Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.

Quick action. `availability target -> error budget -> controls -> test evidence`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Hybrid On-Prem + Cloud AI Setups

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 142 | TOPIC 140 | WEEK 20

Lab - Serve a Model Across AWS + GCP

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 143 | TOPIC 141 | WEEK 21

What Is Edge AI and Why It Matters

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 144 | TOPIC 142 | WEEK 21

Use Cases: Retail, Healthcare, Smart Cities

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 145 | TOPIC 143 | WEEK 21

Introduction to NVIDIA Jetson Devices

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 146 | TOPIC 144 | WEEK 21

Installing JetPack SDK on Jetson

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 147 | TOPIC 145 | WEEK 21

Edge vs Cloud Tradeoffs for AI Deployment

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 148 | TOPIC 146 | WEEK 21

ONNX Runtime for Edge Models

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 149 | TOPIC 147 | WEEK 21

Lab - Deploy YOLOv5 on Jetson Nano

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 150 | TOPIC 148 | WEEK 22

Quantization for Edge Efficiency

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 151 | TOPIC 149 | WEEK 22

Pruning and Model Compression Basics

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 152 | TOPIC 150 | WEEK 22

TensorRT for Edge Inference

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 153 | TOPIC 151 | WEEK 22

Running Vision Models on Raspberry Pi

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 154 | TOPIC 152 | WEEK 22

TinyML and Microcontrollers for AI

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 155 | TOPIC 153 | WEEK 22

Benchmarking Models on Edge Hardware

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 156 | TOPIC 154 | WEEK 22

Lab - Optimize a Model with TensorRT

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 157 | TOPIC 155 | WEEK 23

Why Mobile AI Is Booming

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 158 | TOPIC 156 | WEEK 23

Core ML for iOS - Basics

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 159 | TOPIC 157 | WEEK 23

TensorFlow Lite for Android - Basics

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Deploying On-Device NLP Models

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Power Management in Mobile AI

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Privacy Considerations for On-Device AI

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. `who can do what to which resource under which condition?`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 163 | TOPIC 161 | WEEK 23

Lab - Build a Mobile AI App with TFLite

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 164 | TOPIC 162 | WEEK 24

Introduction to ETL and ELT Pipelines

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. `Record: input -> transform -> output -> checkpoint -> alert`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 165 | TOPIC 163 | WEEK 24

Apache Airflow for AI Workflows

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. `Record: input -> transform -> output -> checkpoint -> alert`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Streaming Data with Apache Kafka

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. `Record: input -> transform -> output -> checkpoint -> alert`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Feature Stores for ML (Feast, Tecton)

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Real-Time Data Preprocessing at Scale

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Data Quality Monitoring for AI Systems

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Build a Streaming Pipeline with Kafka

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. Record: `input -> transform -> output -> checkpoint -> alert`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Infrastructure Challenges of Large Language Models

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 172 | TOPIC 170 | WEEK 25

Memory and Storage Needs of LLMs

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 173 | TOPIC 171 | WEEK 25

Vector Databases - FAISS, Pinecone, Weaviate

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 174 | TOPIC 172 | WEEK 25

RAG (Retrieval-Augmented Generation) Pipelines

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Caching Strategies for LLMs

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Serving LLMs in Production

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deploy a Simple RAG Pipeline

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

DeepSpeed and ZeRO Optimization for LLMs

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. `efficiency = single_device_time / (devices * distributed_time)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Megatron-LM for Large Model Training

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Flash Attention and Memory Optimizations

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Multi-Node Distributed Training for LLMs

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Fine-Tuning vs Parameter-Efficient Tuning

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. $\text{requirement} \rightarrow \text{option} \rightarrow \text{tradeoff} \rightarrow \text{test} \rightarrow \text{decision record}$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Cost Challenges of Generative AI Training

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. $\text{unit cost} = \text{total run cost} / \text{successful useful outputs}$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Fine-Tune a Small LLM with PEFT

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Image and Video Data Challenges in AI Infra

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Training Pipelines for CV Models

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 187 | TOPIC 185 | WEEK 27

EfficientNet and Model Scaling Tradeoffs

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 188 | TOPIC 186 | WEEK 27

Serving Real-Time Video Inference

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. camera/data -> decode -> preprocess -> infer -> postprocess -> sink

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 189 | TOPIC 187 | WEEK 27

DeepStream SDK for Video Analytics

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. Record: input -> transform -> output -> checkpoint -> alert

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Edge-to-Cloud Video Processing

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deploy Real-Time Object Detection

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

NLP Workloads: Tokenization, Embeddings, Transformers

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Training Large Transformer Models

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Infrastructure for BERT and GPT

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Efficient Serving of Embedding Models

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Latency Reduction in NLP Inference

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Deploying Multilingual Models at Scale

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deploy a BERT Model with FastAPI

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

What Is Multimodal AI?

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Handling Text + Image Pipelines

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Training and Serving CLIP Models

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 202 | TOPIC 200 | WEEK 29

Infrastructure for Speech + Text Models

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 203 | TOPIC 201 | WEEK 29

Deploying Video + Text Search Systems

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 204 | TOPIC 202 | WEEK 29

Challenges of Multimodal Model Serving

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deploy CLIP for Image Search

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Basics of Reinforcement Learning Workloads

Core idea. Balance simulation or environment throughput, experience transport, learner updates, policy freshness, and safety validation.

Evidence check. Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Quick action. `actors -> experience -> replay -> learner -> policy distribution`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Simulation Environments for RL (Gym, Isaac)

Core idea. Balance simulation or environment throughput, experience transport, learner updates, policy freshness, and safety validation.

Evidence check. Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Quick action. `actors -> experience -> replay -> learner -> policy distribution`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 208 | TOPIC 206 | WEEK 30

Distributed RL Training Infrastructures

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 209 | TOPIC 207 | WEEK 30

Real-Time Serving of RL Agents

Core idea. Expose a versioned prediction contract with deterministic preprocessing, health semantics, capacity policy, and measurable SLOs.

Evidence check. Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Quick action. `curl -fsS http://localhost:8000/health`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 210 | TOPIC 208 | WEEK 30

Scaling RL for Robotics

Core idea. Balance simulation or environment throughput, experience transport, learner updates, policy freshness, and safety validation.

Evidence check. Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Quick action. `actors -> experience -> replay -> learner -> policy distribution`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Infrastructure for Online Learning Agents

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Train RL Agent with Ray RLlib

Core idea. Balance simulation or environment throughput, experience transport, learner updates, policy freshness, and safety validation.

Evidence check. Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Quick action. `actors -> experience -> replay -> learner -> policy distribution`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

What Is Large-Scale Training?

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Data Parallelism vs Model Parallelism Revisited

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Pipeline Parallelism in Transformers

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Distributed Optimizers and Gradient Sync

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Infrastructure Bottlenecks in Training LLMs

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Fault Tolerance in Multi-Node Training

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. `efficiency = single_device_time / (devices * distributed_time)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Train Transformer Across Multiple Nodes

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

DeepSpeed ZeRO-2/3 Optimizations

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Fully Sharded Data Parallel (FSDP)

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. $\text{efficiency} = \text{single_device_time} / (\text{devices} * \text{distributed_time})$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Flash Attention in Large Models

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Checkpointing Strategies for Multi-Node Systems

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. `efficiency = single_device_time / (devices * distributed_time)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Elastic Training with Kubernetes

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Scaling Beyond 1,000 GPUs

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Train with DeepSpeed ZeRO-3

Core idea. Partition work or state only when the memory or elapsed-time benefit exceeds communication and coordination overhead.

Evidence check. Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Quick action. `efficiency = single_device_time / (devices * distributed_time)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

What Is Enterprise MLOps?

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Introduction to Kubeflow

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Introduction to MLflow at Scale

Core idea. Tie each result to code, data, environment, parameters, metrics, artifacts, and an accountable lifecycle decision.

Evidence check. Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Quick action. `mlflow server --host 127.0.0.1 --port 5000`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

SageMaker Pipelines - Basics

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

GCP Vertex AI Pipelines

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 232 | TOPIC 230 | WEEK 33

Azure ML Pipelines

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 233 | TOPIC 231 | WEEK 33

Lab - Build a Pipeline in Kubeflow

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 234 | TOPIC 232 | WEEK 34

Model Registry in Enterprise MLOps

Core idea. Tie each result to code, data, environment, parameters, metrics, artifacts, and an accountable lifecycle decision.

Evidence check. Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Quick action. `mlflow server --host 127.0.0.1 --port 5000`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Continuous Training (CT) Pipelines

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Automating Drift Retraining with Kubeflow

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Feature Store Integration in Pipelines

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Governance in Enterprise MLOps

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Audit Trails and Compliance Logging

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. `who can do what to which resource under which condition?`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Automate Drift Retraining with Kubeflow

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

What Is Model Optimization for Infra Efficiency?

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Quantization Basics

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. baseline -> transform -> calibrate/train -> validate -> benchmark

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Pruning Basics

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. baseline -> transform -> calibrate/train -> validate -> benchmark

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Distillation Basics

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Structured vs Unstructured Sparsity

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Benchmarking Optimized Models

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Quantize a Vision Model

Core idea. Budget media decode, preprocessing, transfer, batching, inference, and postprocessing as one end-to-end pipeline.

Evidence check. Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Quick action. `camera/data -> decode -> preprocess -> infer -> postprocess -> sink`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Mixed Precision Training with AMP

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Quantization-Aware Training (QAT)

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Advanced Distillation Techniques

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Sparse Training and Hardware Impacts

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Compiler Optimizations (XLA, TorchDynamo)

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. `baseline -> transform -> calibrate/train -> validate -> benchmark`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Infra Tradeoffs: Accuracy vs Efficiency

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Train with Mixed Precision

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. baseline -> transform -> calibrate/train -> validate -> benchmark

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

What Is Federated Learning?

Core idea. Start with a threat model, then choose privacy controls while measuring utility, compute, communication, and operational complexity.

Evidence check. Record trust assumptions, privacy budget or protocol, client skew, dropout behavior, aggregation integrity, and utility loss.

Quick action. threat -> control -> residual risk -> utility cost -> evidence

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Privacy-Preserving AI at Scale

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Federated Learning with TensorFlow Federated

Core idea. Start with a threat model, then choose privacy controls while measuring utility, compute, communication, and operational complexity.

Evidence check. Record trust assumptions, privacy budget or protocol, client skew, dropout behavior, aggregation integrity, and utility loss.

Quick action. threat -> control -> residual risk -> utility cost -> evidence

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Secure Aggregation Protocols

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Federated Data Challenges

Core idea. Start with a threat model, then choose privacy controls while measuring utility, compute, communication, and operational complexity.

Evidence check. Record trust assumptions, privacy budget or protocol, client skew, dropout behavior, aggregation integrity, and utility loss.

Quick action. `threat -> control -> residual risk -> utility cost -> evidence`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Edge Deployment of Federated Models

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Train a Federated Model with TFF

Core idea. Start with a threat model, then choose privacy controls while measuring utility, compute, communication, and operational complexity.

Evidence check. Record trust assumptions, privacy budget or protocol, client skew, dropout behavior, aggregation integrity, and utility loss.

Quick action. `threat -> control -> residual risk -> utility cost -> evidence`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Why Privacy Matters in AI Infra

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Homomorphic Encryption Basics

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Differential Privacy for AI Models

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Secure Multi-Party Computation (MPC)

Core idea. Start with a threat model, then choose privacy controls while measuring utility, compute, communication, and operational complexity.

Evidence check. Record trust assumptions, privacy budget or protocol, client skew, dropout behavior, aggregation integrity, and utility loss.

Quick action. threat -> control -> residual risk -> utility cost -> evidence

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Tradeoffs in Privacy-Preserving AI

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Industry Applications of Privacy-Preserving AI

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Apply Differential Privacy in Training

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Attacks on AI Infrastructure

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Model Poisoning Attacks

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Data Poisoning Attacks

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Membership Inference Attacks

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Adversarial Examples in Deployment

Core idea. Trade numerical precision or computation for efficiency only within an explicit quality and hardware-support envelope.

Evidence check. Compare baseline and optimized accuracy, latency, throughput, memory, artifact size, energy, and unsupported operators.

Quick action. baseline -> transform -> calibrate/train -> validate -> benchmark

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Mitigation Strategies for Infra Security

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Defend Against Adversarial Attacks

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

What Is Multi-Tenancy?

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Resource Sharing Across Teams

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Cost Allocation for Multi-Tenant Infra

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. $\text{unit cost} = \text{total run cost} / \text{successful useful outputs}$

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Role-Based Access Control (RBAC)

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. who can do what to which resource under which condition?

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Isolation Strategies in AI Systems

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Monitoring Multi-Tenant Environments

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Configure Multi-Tenant Cluster

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Challenges Startups Face in AI Infra

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lean GPU Cloud Solutions

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Open-Source MLOps Tools for Startups

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Budget Optimization for Small Teams

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Scaling from MVP to Production AI

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Vendor Lock-In Risks for Startups

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deploy AI Infra on Low Budget

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Enterprise-Scale Infra Needs

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Vendor Selection for AI Infrastructure

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 292 | TOPIC 290 | WEEK 42

Hybrid On-Prem + Cloud AI Strategy

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 293 | TOPIC 291 | WEEK 42

Compliance and Regulatory Burdens

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. `who can do what to which resource under which condition?`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 294 | TOPIC 292 | WEEK 42

Integration with Enterprise IT Systems

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Scaling Teams for AI Infra Management

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Design Enterprise AI Infra Plan

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Real-Time AI Use Cases (Ads, Fraud, Personalization)

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Latency Challenges in Real-Time AI

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Streaming Infrastructure for Real-Time AI

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. Record: input -> transform -> output -> checkpoint -> alert

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Deploying Low-Latency APIs

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Scaling Real-Time Recommendation Systems

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Cost Challenges in Real-Time AI

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Build Real-Time Fraud Detection Pipeline

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

AI Infra in Self-Driving Cars

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Robotics AI Infrastructure Basics

Core idea. Balance simulation or environment throughput, experience transport, learner updates, policy freshness, and safety validation.

Evidence check. Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Quick action. `actors -> experience -> replay -> learner -> policy distribution`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Sensor Fusion Data Pipelines

Core idea. Make data movement replayable and observable with schemas, idempotency, checkpoints, backpressure, and explicit ownership.

Evidence check. Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Quick action. `Record: input -> transform -> output -> checkpoint -> alert`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Real-Time AI in Safety-Critical Systems

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Simulation Infrastructure for Autonomous Vehicles

Core idea. Balance simulation or environment throughput, experience transport, learner updates, policy freshness, and safety validation.

Evidence check. Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Quick action. `actors -> experience -> replay -> learner -> policy distribution`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Edge Deployment in Robotics

Core idea. Design to the device envelope: memory, compute, energy, thermals, runtime support, connectivity, and updateability.

Evidence check. Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Quick action. `measure on target hardware - desktop results are not a proxy`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deploy AI Agent for Robotics Simulation

Core idea. Balance simulation or environment throughput, experience transport, learner updates, policy freshness, and safety validation.

Evidence check. Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Quick action. `actors -> experience -> replay -> learner -> policy distribution`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Case Study: OpenAI Infra for GPT

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Case Study: Google Infra for DeepMind

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 313 | TOPIC 311 | WEEK 45

Case Study: Meta Infra for LLaMA

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 314 | TOPIC 312 | WEEK 45

Case Study: Tesla AI Infra for FSD

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 315 | TOPIC 313 | WEEK 45

Case Study: Netflix AI Infra for Recommendations

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 316 | TOPIC 314 | WEEK 45

Case Study: Healthcare AI Infra at Scale

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 317 | TOPIC 315 | WEEK 45

Lab - Analyze Case Study Infra Architecture

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 318 | TOPIC 316 | WEEK 46

Trends in AI Chips (GPUs, TPUs, NPUs)

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Cloud Evolution for AI

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

AI + Quantum Computing Infrastructure

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Software Trends in AI Infra (Rust, Mojo)

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Green AI and Sustainable Infrastructure

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Global AI Regulation Impact on Infra

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Design Future-Proof AI Infra

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 325 | TOPIC 323 | WEEK 47

Review of Hardware Concepts

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 326 | TOPIC 324 | WEEK 47

Review of Cloud Infrastructure

Core idea. Choose managed and self-managed cloud building blocks from workload requirements, quotas, failure domains, governance, and exit cost.

Evidence check. Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Quick action. `record region + instance + accelerator + image + storage + hourly estimate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 327 | TOPIC 325 | WEEK 47

Review of Containerization and Kubernetes

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 328 | TOPIC 326 | WEEK 47

Review of MLOps Pipelines

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 329 | TOPIC 327 | WEEK 47

Review of Monitoring and Security

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 330 | TOPIC 328 | WEEK 47

Review of Cost Optimization Strategies

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Mini-Project Review Sprint

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Choosing a Capstone Domain (NLP, Vision, Generative AI)

Core idea. Treat retrieval, context assembly, model execution, caching, evaluation, and safety as separate measurable stages.

Evidence check. Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Quick action. `request -> retrieve -> rerank -> assemble -> generate -> evaluate`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Defining Success Metrics for Infra Project

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Designing Initial Infrastructure Blueprint

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Estimating Hardware and Cloud Costs

Core idea. Optimize cost per useful, reliable outcome by improving utilization and eliminating idle, retry, storage, and transfer waste.

Evidence check. Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Quick action. `unit cost = total run cost / successful useful outputs`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Selecting Tools and Frameworks for Build

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Identifying Risks and Mitigation Plans

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Capstone Project Proposal

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Setting Up Base Cloud/GPU Environment

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Containerizing Models for Deployment

Core idea. Package code and runtime dependencies into a repeatable image with a narrow, explicit execution contract.

Evidence check. Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Quick action. `docker build -t ai-app:local . && docker run --rm ai-app:local`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Setting Up Orchestration with Kubernetes

Core idea. Declare desired state, resources, health, placement, storage, and rollout behavior so the control plane can reconcile safely.

Evidence check. Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Quick action. `kubectl get pods -A -o wide && kubectl top pods -A`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Configuring Storage and Data Pipelines

Core idea. Select storage by access pattern, consistency, durability, scale, recovery, and cost rather than by familiarity.

Evidence check. Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Quick action. `du -sh .; find . -type f | wc -l`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Implementing CI/CD for Model Deployment

Core idea. Automate a gated path from change to verified artifact to controlled rollout, with rollback as a tested operation.

Evidence check. Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Quick action. `Build -> test -> scan -> sign -> deploy -> verify -> promote/rollback`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Building Initial Monitoring Dashboards

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deploy First Working Infra Prototype

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Scaling Training Across Multi-GPU Nodes

Core idea. Match compute kernels, memory capacity, memory bandwidth, precision support, and interconnect topology to the workload.

Evidence check. Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Quick action. `nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total --format=csv`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Implementing Drift Detection Pipeline

Core idea. Compare live data and outcomes with a justified reference, then route meaningful change into diagnosis and response.

Evidence check. Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.

Quick action. `reference window + current window + threshold + owner + action`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Securing Model Endpoints with IAM

Core idea. Reduce exposure through verified identity, least privilege, segmented trust, protected data and artifacts, and auditable policy.

Evidence check. Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Quick action. `who can do what to which resource under which condition?`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Cost Monitoring and Optimization Setup

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Configuring High Availability Clusters

Core idea. Turn failure assumptions into redundancy, bounded recovery objectives, tested failover, and graceful degradation.

Evidence check. Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.

Quick action. `availability target -> error budget -> controls -> test evidence`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Stress Testing the AI System

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Deliver Scalable Infra Deployment

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Conducting End-to-End Testing

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Adding Redundancy and Failover Mechanisms

Core idea. Turn failure assumptions into redundancy, bounded recovery objectives, tested failover, and graceful degradation.

Evidence check. Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.

Quick action. availability target -> error budget -> controls -> test evidence

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Polishing Monitoring and Alerting Dashboards

Core idea. Connect user-visible service health to model behavior and infrastructure saturation with bounded-cardinality telemetry.

Evidence check. Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Quick action. `rate(requests_total[5m]); histogram_quantile(0.99, ...)`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Generating Documentation for AI Infra

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Creating Demo API Endpoints

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Preparing Final Capstone Report

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Lab - Capstone Infra Walkthrough

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Preparing Final Demo Environment

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Recording Metrics and Performance Benchmarks

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Building Visuals of Infra Architecture

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. requirement -> option -> tradeoff -> test -> decision record

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Delivering Stakeholder Presentation

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. claim -> metric -> method -> result -> decision

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 364 | TOPIC 362 | WEEK 52

Peer Review and Feedback on Projects

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 365 | TOPIC 363 | WEEK 52

Reflection on Zero-to-Hero Journey

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

LECTURE 366 | TOPIC 364 | WEEK 52

Future Learning Paths in AI Infrastructure

Core idea. Translate the topic into an explicit infrastructure decision with requirements, alternatives, tradeoffs, and evidence.

Evidence check. Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Quick action. `requirement -> option -> tradeoff -> test -> decision record`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

Graduation & Certification Showcase

Core idea. Tell a traceable engineering story from problem and constraints to architecture, evidence, cost, risk, and next action.

Evidence check. Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Quick action. `claim -> metric -> method -> result -> decision`

Exit prompt. Explain the decision this topic changes, one failure mode, and the metric that would reveal it.

KEEP THE REFERENCE ALIVE

Tooling evolves. Revalidate commands against official documentation, record the version used in every lab, and prefer repeatable evidence over memorized defaults.

Course coverage audit

Scope	Count
Introductory platform lectures	2
Numbered curriculum topics	365
Total cards	367
Course weeks	52